

Cloud Computing (COM467)

Unit 1

Introduction to Cloud Computing

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2025

- Evolution of Cloud Computing
- Characteristics of Cloud Computing
- Types of Cloud and its Cloud Services
- Benefits and Challenges of Cloud Computing
- Applications Cloud Computing
- Cloud Storage
- Cloud Services Requirements,
- Cloud and Dynamic Infrastructure
- Cloud Adoption

CLOUD

- The term "**Cloud**" refers to a network or the Internet, representing resources available at a remote location.
- The **cloud** is a network of remote servers hosted on the Internet to **store, manage, and process data**, instead of using a local server or personal computer.
- It allows access to **servers, software, and databases** over the Internet.
- Cloud services can operate over **public or private networks** such as WAN, LAN, or VPN.
- Common applications include **email, web conferencing, and customer relationship management (CRM)**.

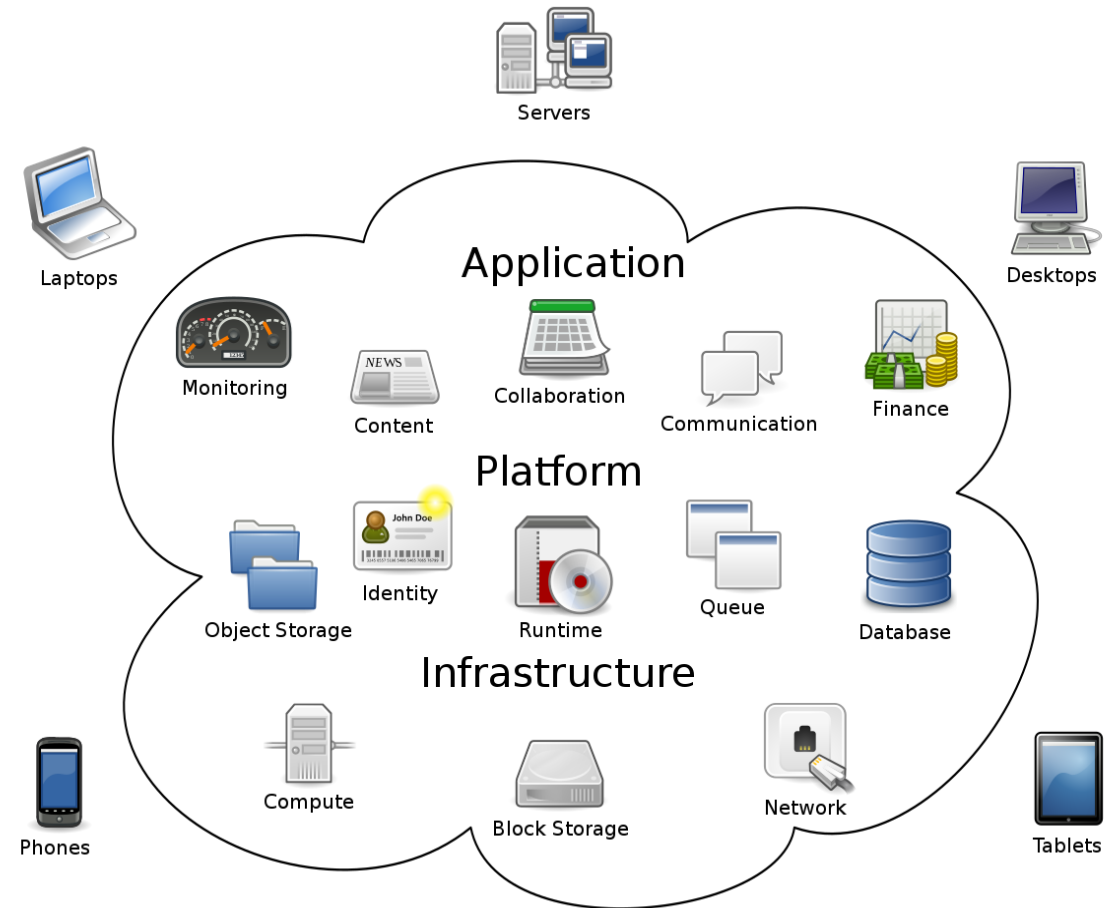
CLOUD COMPUTING

- Simply put, Cloud Computing is the delivery of IT services and resources on-demand via the Internet.
- **Cloud computing** is a model that enables **on-demand delivery of computing resources**—such as servers, storage, and applications—**over the Internet**.
- It allows users to **access resources anytime and anywhere**, similar to how we use utilities like **electricity or water**.
- It refers to manipulating, configuring and accessing the hardware and software resources remotely.
- Instead of buying and maintaining hardware or software, users **rent resources from cloud providers** when needed, ensuring flexibility and cost savings.
- This model eliminates the need for users to invest heavily in IT infrastructure and personnel.
- Cloud computing as a model of computing shifts IT resources from being **locally owned and managed** to being **delivered as a service** over the internet.
- It offers online data storage, infrastructure and applications.

CLOUD COMPUTING

- To help clarify how cloud computing has changed the nature of commercial system deployment, consider these three examples:
 - **Google** - In the last decade, Google has built a worldwide network of datacenters to service its search engine. In doing so Google has captured a substantial portion of the world's advertising revenue. That revenue has enabled Google to offer free software to users based on that infrastructure and has changed the market for user-facing software. This is the classic **Software as a Service**.
 - **Azure Platform** - By contrast, Microsoft is creating the Azure Platform. It enables .NET Framework applications to run over the Internet as an alternate **platform** for Microsoft developer software running on desktops.
 - **Amazon Web Services** - One of the most successful cloud-based businesses is Amazon Web Services, which is an **Infrastructure as a Service** offering that lets you rent virtual computers on Amazon's own infrastructure.

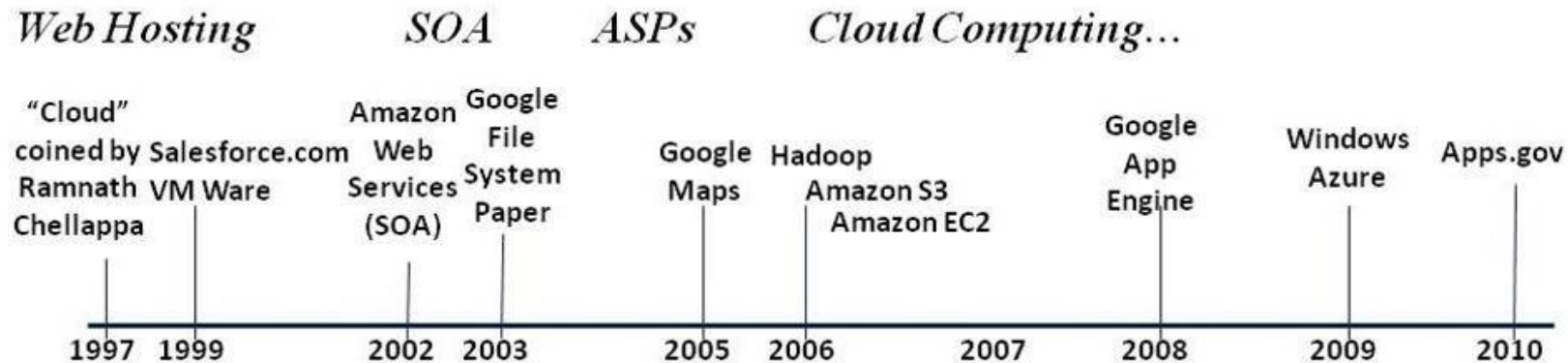
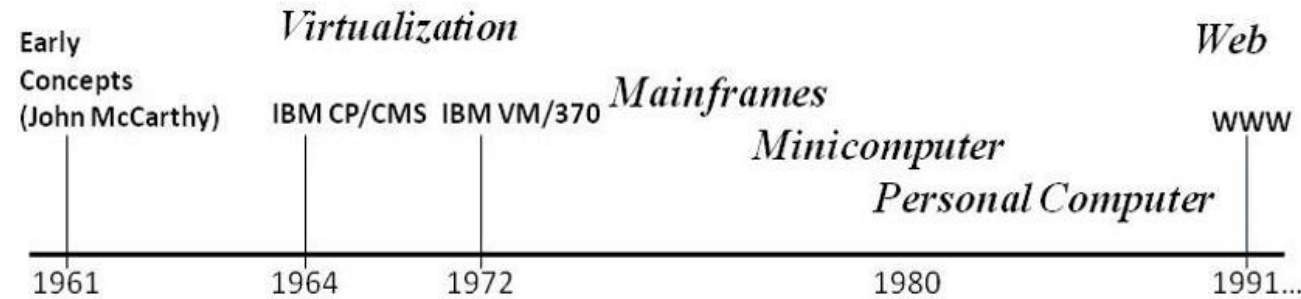
CLOUD COMPUTING



CLOUD EXAMPLES

- **Google Drive / Dropbox** – Storing files online so you can access them from any device without saving them on your computer.
- **Microsoft Office 365 / Google Docs** – Using word processors, spreadsheets, or presentation tools directly in the cloud, allowing real-time collaboration.
- **Amazon Web Services (AWS) / Microsoft Azure** – Businesses using cloud servers to host websites, applications, or databases instead of maintaining physical servers.
- **Netflix / Spotify** – Streaming videos or music from cloud servers without downloading the full content to your device.
- **Zoom / Google Meet** – Conducting video calls and meetings using cloud-based software without installing complex infrastructure.

EVOLUTION OF CLOUD COMPUTING



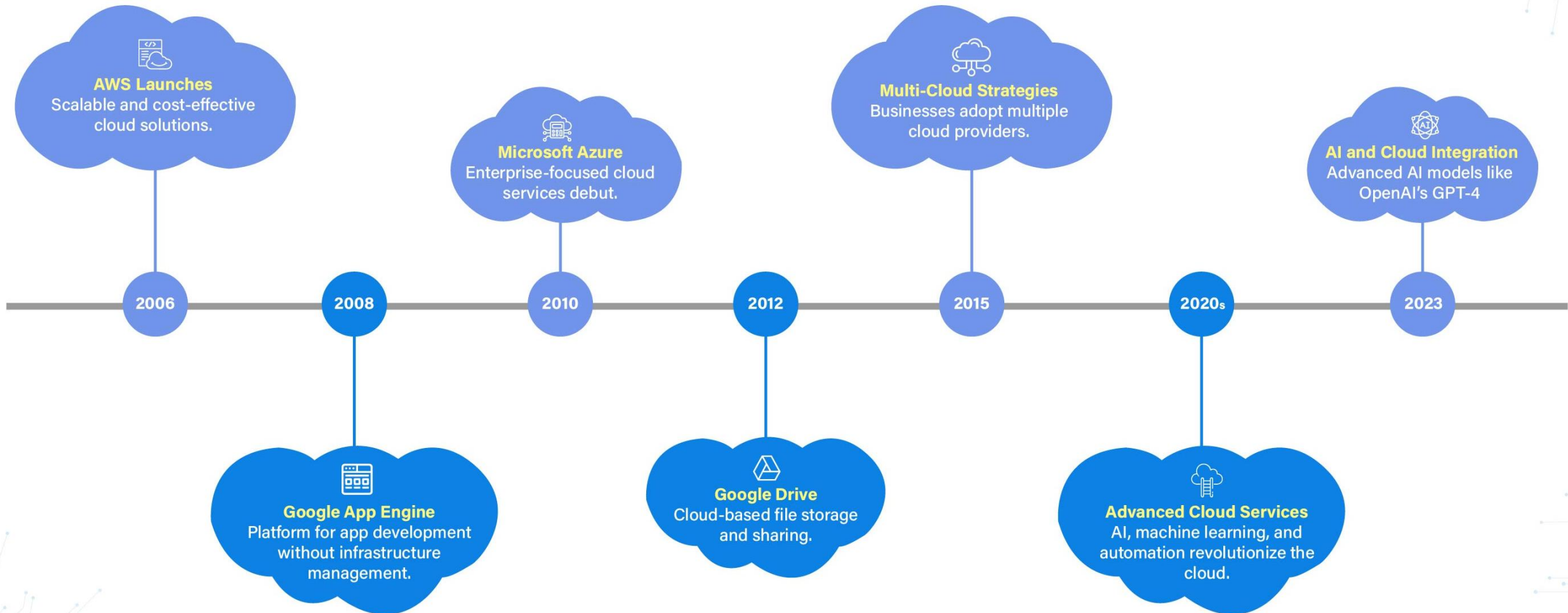
EVOLUTION OF CLOUD COMPUTING

- Cloud computing is all about renting computing services.
- **1950s:** Concept emerged with **mainframe computing**, mainframe computers accessed via thin/static clients.
- **1960s: Utility Computing** idea proposed by *John McCarthy* — “computation may someday be organized as a public utility.”
- **1980s:** Rise of **Grid Computing** (On-Demand Computing) — distributes large tasks across multiple servers for parallel processing.
- **1990s: Virtualization** extended beyond servers to storage, network, and applications, enabling flexible, infrastructure-independent platforms.
- **Core Technologies Behind Cloud Computing**
 - Grid Computing → Resource sharing and workload distribution.
 - Virtualization → Abstracts hardware resources.
 - Utility Computing → Pay-per-use business model.
 - SaaS (Software as a Service) → Application-level service delivery.

EVOLUTION OF CLOUD COMPUTING

- **1999:** Salesforce launched — first major SaaS (Software as a Service) model on the web.
- **2002:** Amazon Web Services (AWS) started offering cloud-based storage and computation.
- **2006:** Amazon launched Elastic Compute Cloud (EC2), marking the start of modern **cloud infrastructure (IaaS)**.
- **2008:** Google App Engine introduced PaaS (Platform as a Service).
- **2010:** Microsoft Azure and other major cloud platforms entered the market.
- **2020s:** Cloud evolved into **hybrid, edge, and serverless computing**, integrating **AI, IoT, and big data**.

Evolution of Cloud Computing



MAINFRAME COMPUTING

- Mainframe computing refers to the use of large, powerful, centralized computers (mainframes) to process and manage vast amounts of data and transactions simultaneously.
- All processing happens in a single system, and multiple users access it through **terminals** (thin clients).
- IBM zSeries Mainframes used by banks and airlines for transaction processing (e.g., ATM transactions, flight reservations).
- A bank's core system handling millions of account updates per second.

DISTRIBUTED COMPUTING

- Distributed computing involves a **collection of independent computers** working together as a single system to solve a common problem.
- Tasks are divided and executed in parallel across multiple machines.
- **The key features are:** Decentralized processing, Resource sharing over a network, Fault tolerance.
- **Example: Apache Hadoop**, which distributes big data processing across many nodes.

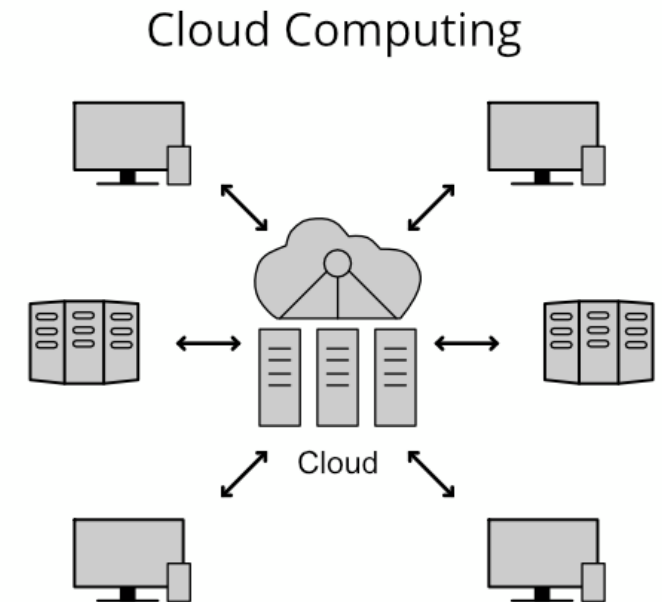
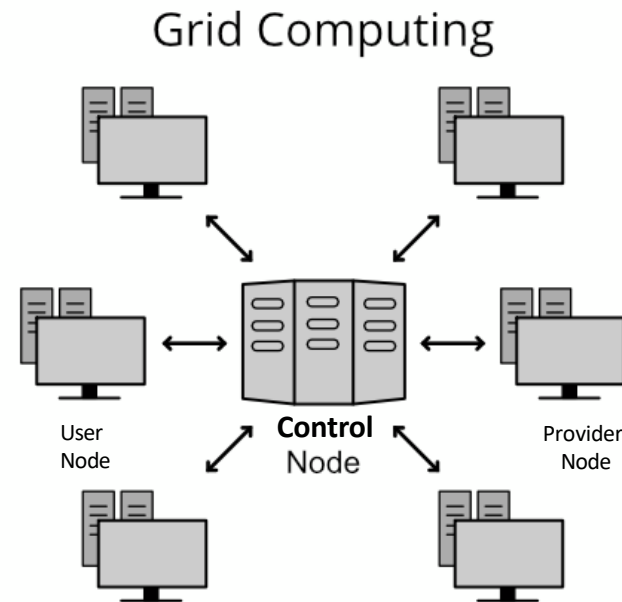
CLUSTER COMPUTING

- Cluster computing connects **multiple computers (nodes)** in a local network to work as a single, high-performance system.
- All nodes are typically identical and located in the same place.
- **The main features are:**
 - Nodes connected by high-speed LAN
 - Centralized job scheduling
 - High availability and performance
- **Example:**

Google's early search engine clusters used thousands of commodity PCs to index the web.

GRID COMPUTING

- It is a distributed computing resource to accomplish a common goal.
- It is connected by parallel nodes that form a computer cluster and runs on an operating system. It is a distributed system with non-interactive workloads including a large number of files.
- The cluster can be either a small work station or a large network. Its common uses are ATM banking, back-end infrastructure, and scientific marketing research.
- We can make grid computing with the help of the application, which can use for computation problem. That are connected in a parallel networking environment.



CLUSTER COMPUTING VS GRID COMPUTING

Basis	Cluster Computing	Grid Computing
Definition	A group of closely connected computers (nodes) working together as a single system .	A distributed network of independent computers that work together to perform large-scale tasks.
Location	Nodes are located in the same physical location (e.g., a data center or lab).	Systems are geographically distributed , possibly across multiple organizations or regions.
Hardware & OS	All nodes usually have similar hardware and operating systems .	Nodes can have different hardware, operating systems, and configurations .
Network Type	Uses high-speed Local Area Network (LAN) connections.	Uses Wide Area Network (WAN) or the Internet for communication.
Resource Sharing	Resources are tightly coupled and managed by a single organization .	Resources are loosely coupled , often shared among multiple organizations .
Scalability	Limited scalability – confined to local infrastructure.	Highly scalable – can include thousands of nodes globally.
Purpose	Focused on high performance computing (HPC) and parallel processing .	Focused on resource sharing, collaboration, and distributed problem solving .
Management	Managed as a single system with centralized scheduling.	Managed in a decentralized way using grid middleware.
Example	A university's local HPC cluster for running simulations.	SETI@home or BOINC projects using computers worldwide for research.

UTILITY COMPUTING

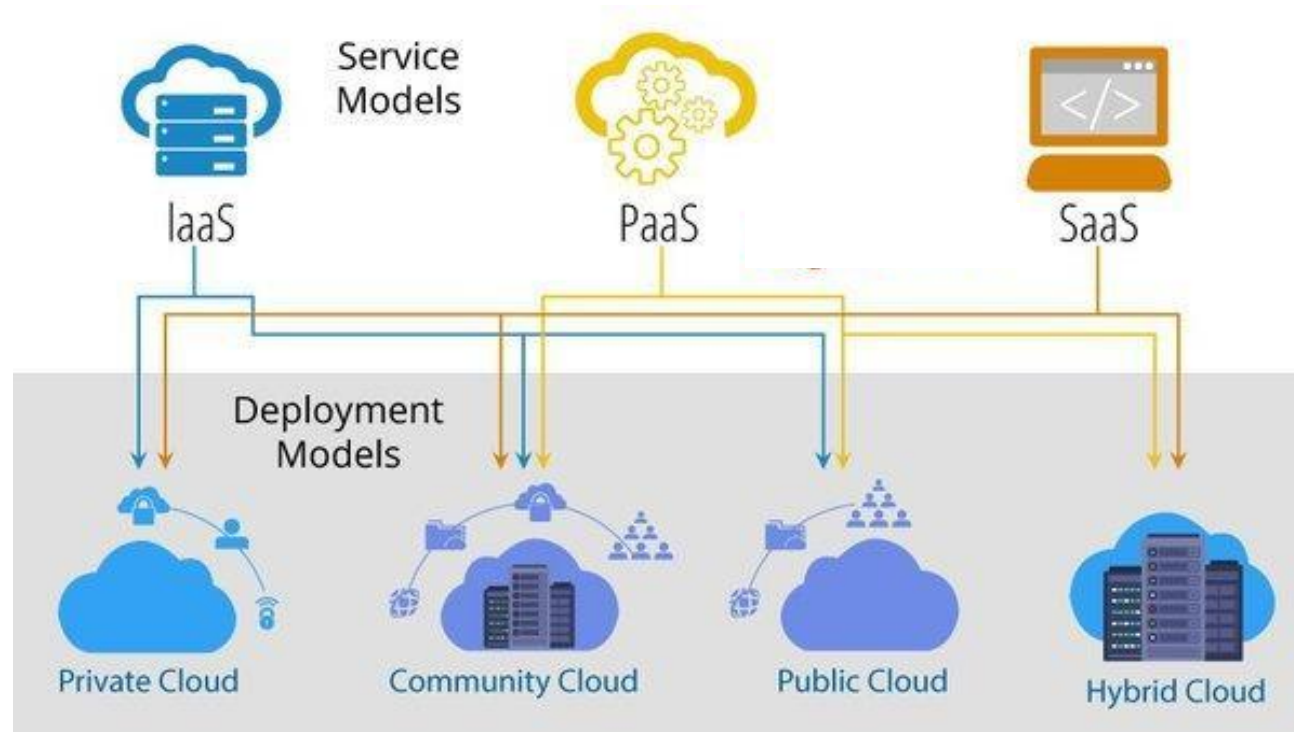
- Utility computing provides computing resources (like CPU, storage, or network) as a **metered service**, just like utilities such as electricity or water.
- Utility computing is a service provisioning model.
- This service provisioning modeling provides computing resources and infrastructure management to the customer as per their demand.
- The customers are charged for them as you go basis without any upfront cost.
- The utility model maximizes the efficient use of resources while minimizing the associated cost.
- Utility computing has an advantage that there will be a low initial cost to acquire computer resources.
- The customer can access the infinite amount of computing solution with the help of the internet or a virtual private network.
- A provider can perform, the backend infrastructure and computing resources management.
- Examples:
 - Amazon EC2 (Elastic Compute Cloud) offering virtual servers billed per hour.
 - Google Cloud Storage charging per GB stored.

CHARACTERISTICS OF CLOUD COMPUTING

No.	Characteristic	Description
1	On-demand self-service	Users can automatically provision computing resources (e.g., server time, storage) without human interaction with the provider.
2	Broad network access	Services are available over the network and accessed through standard mechanisms (e.g., web browsers, mobile apps).
3	Resource pooling	Resources (e.g., storage, memory, processing) are pooled to serve multiple users using a multi-tenant model.
4	Rapid elasticity	Resources can be quickly scaled up or down based on demand — appearing to be unlimited.
5	Measured service	Resource usage is monitored, controlled, and reported — enabling pay-as-you-go billing.

TYPES OF CLOUD AND IT'S SERVICE MODELS

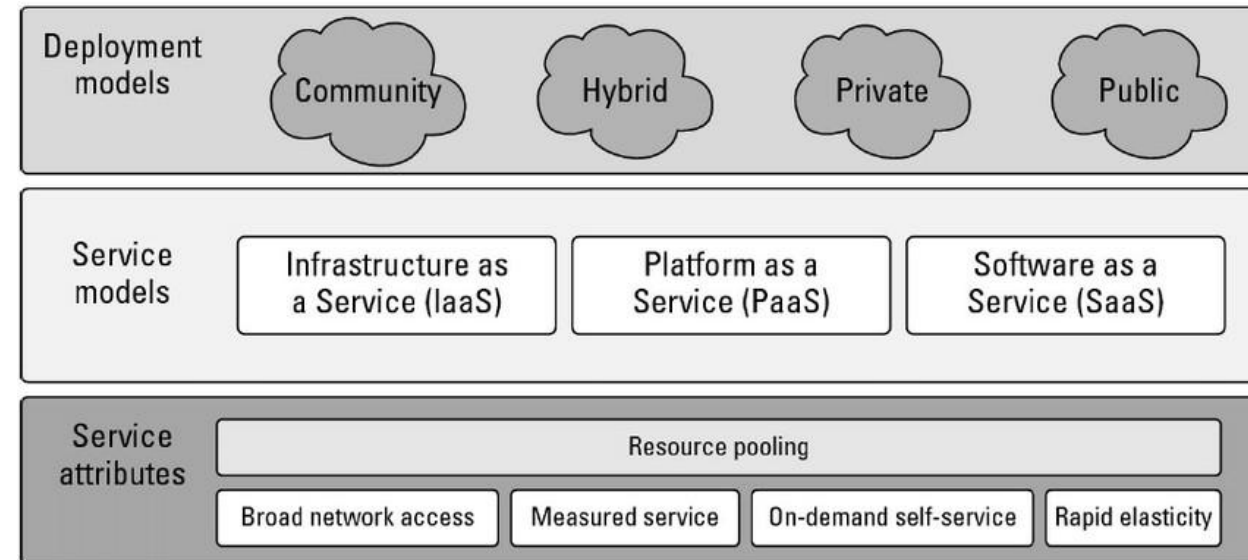
- Most people separate cloud computing into two distinct sets of models:
 1. **Deployment Models:** This refers to the **location** and **management** of the cloud's infrastructure.
 2. **Service Models:** This consists of the **particular types of services** that you can access on a cloud computing platform.



TYPES OF CLOUD AND IT'S SERVICE MODELS

The NIST model

- Defined by The U.S. National Institute of Standards and Technology (NIST) (2011)
- Model includes:
 - **5 Essential Characteristics**
 - **3 Service Models (IaaS, PaaS, SaaS)**
 - **4 Deployment Models (Public, Private, Hybrid, Community)**
- It provides a **standard framework** for understanding and implementing cloud computing technologies.
- The NIST model originally **did not require a cloud to use virtualization to pool resources**, nor did it absolutely require that a cloud support multi-tenancy in the earliest definitions of cloud computing.
- **Multi-tenancy** is the sharing of resources among two or more clients.
- The latest version of the NIST definition **does require that cloud computing networks** use virtualization and support multi-tenancy.

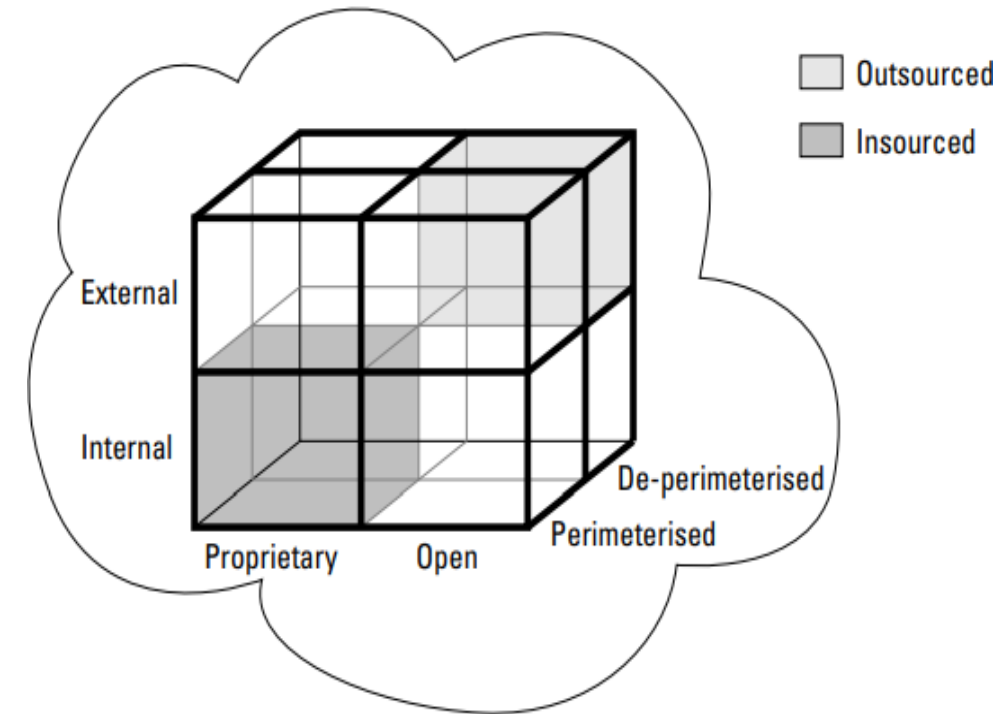


TYPES OF CLOUD AND IT'S SERVICE MODELS

The Cloud Cube Model

- Developed by the **Jericho Forum (2009)** to help organizations **understand different types of cloud environments** and **choose the right deployment strategy** based on **security, ownership, and location** of data.
- The **Open Group** maintains an association called the **Jericho** whose main focus is how to protect cloud networks.
- It categorizes a cloud network based on four dimensional factors.
- The four dimensions of the Cloud Cube Model:
 - **Physical Location of the Data:** Internal (I) / External (E) determines our organization's boundaries.
 - **Ownership:** Proprietary (P) / Open (O) is a measure of not only the technology ownership, but of interoperability, ease of data transfer, and degree of vendor application lock-in.
 - **Security Boundary:** Perimeterised (Per) / De- perimeterised (D-p) is a measure of whether the operation is inside or outside the security boundary or network firewall.
 - **Sourcing:** Insourced or Outsourced means whether the service is provided by the customer or the service provider.

The Jericho Forum's Cloud Cube Model



TYPES OF CLOUD AND IT'S SERVICE MODELS

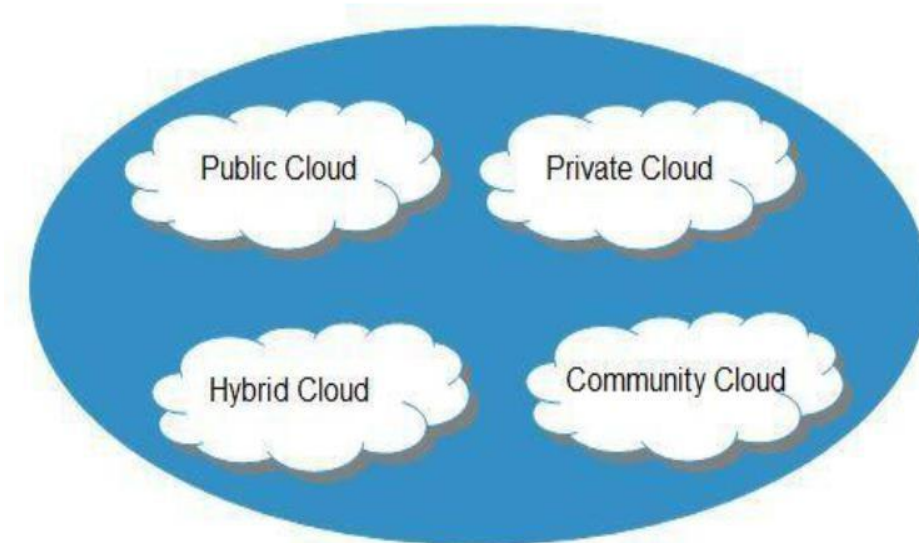
The Cloud Cube Model

Dimension	Options	Meaning
1. Internal / External	Internal, External	Where the cloud services are physically located and who owns the infrastructure.
2. Proprietary / Open	Proprietary, Open	Determines whether the technology and interfaces are closed (vendor-controlled) or open (standard-based).
3. Perimeterised / De-perimeterised	Perimeterised, De-perimeterised	Refers to whether access to cloud resources is controlled by traditional network boundaries (firewalls) or through identity-based and encrypted methods.
4. Insourced / Outsourced	Insourced, Outsourced	Whether the cloud services are managed internally by the organization or by an external service provider.

TYPES OF CLOUD AND IT'S SERVICE MODELS

Deployment Models

- Deployment models define the type of access to the cloud, i.e., how the cloud is located?
- Cloud can have any of the four types of access:
 - **Public Cloud**
 - allows systems and services to be easily accessible to the general public. e.g., e-mail
 - **Private Cloud**
 - allows systems and services to be accessible within an organization
 - **Hybrid cloud (mixture of public and private cloud)**
 - the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.
 - **Community Cloud**
 - allows systems and services to be accessible by group of organizations



DEPLOYMENT MODELS

Model	Description	Example
Private Cloud	Cloud infrastructure operated solely for one organization. Can be on-premises or hosted.	Government or large enterprise cloud
Public Cloud	Cloud infrastructure open to the general public and owned by a service provider.	AWS, Google Cloud, Microsoft Azure
Hybrid Cloud	Combination of private and public clouds that remain distinct but share data and applications.	A company using AWS for public-facing apps and private cloud for internal data
Community Cloud	Shared infrastructure for a specific community with common interests (e.g., education, banking).	Government or university consortium cloud

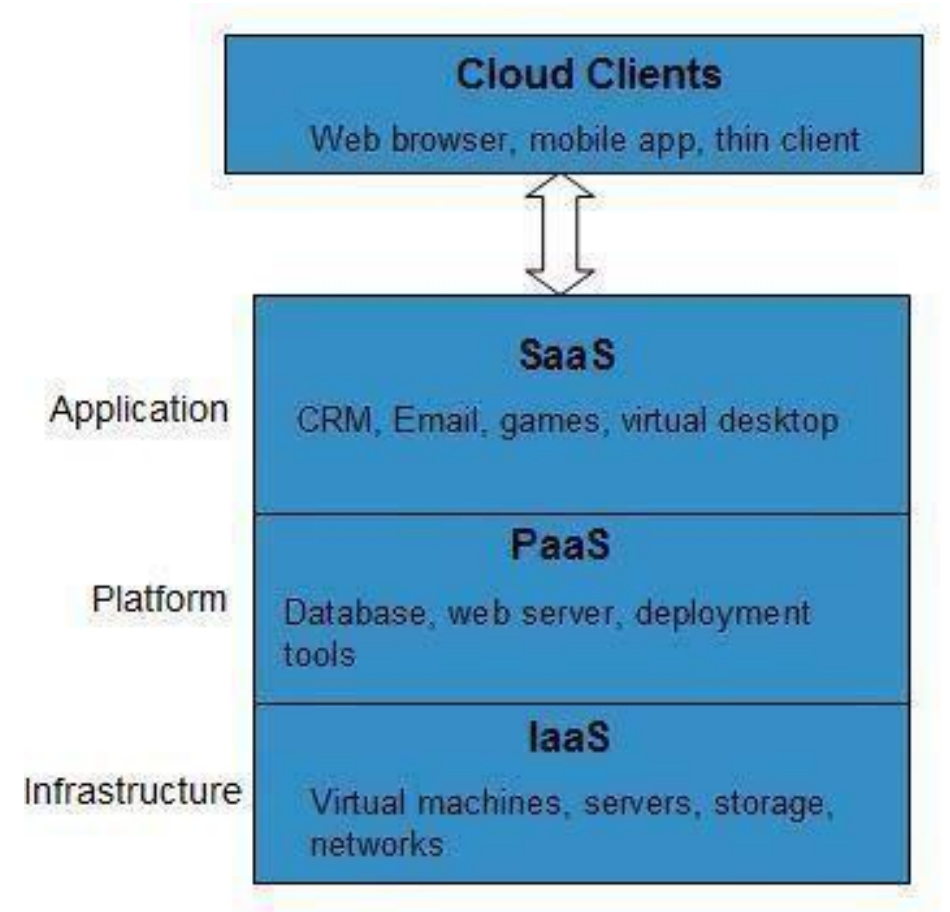
TYPES OF CLOUD AND IT'S SERVICE MODELS

■ Service Models

- Service Models are the reference models on which the Cloud Computing is based.
- These can be categorized into three basic service models :
 - **Infrastructure as a Service (IaaS)**
 - **Platform as a Service (PaaS)**
 - **Software as a Service (SaaS)**
- The **Infrastructure as a Service (IaaS)** is the most basic level of service.
- Each of the service models make use of the underlying service model, i.e., each inherits the security and management mechanism from the underlying model.
- There are many other service models all of which can take the form like **XaaS**, i.e.,
- **Anything as a Service.**
 - This can be Network as a Service (NaaS), Business as a Service (BaaS), Identity as a Service (IaaS), Database as a Service (DaaS)

TYPES OF CLOUD AND IT'S SERVICE MODELS

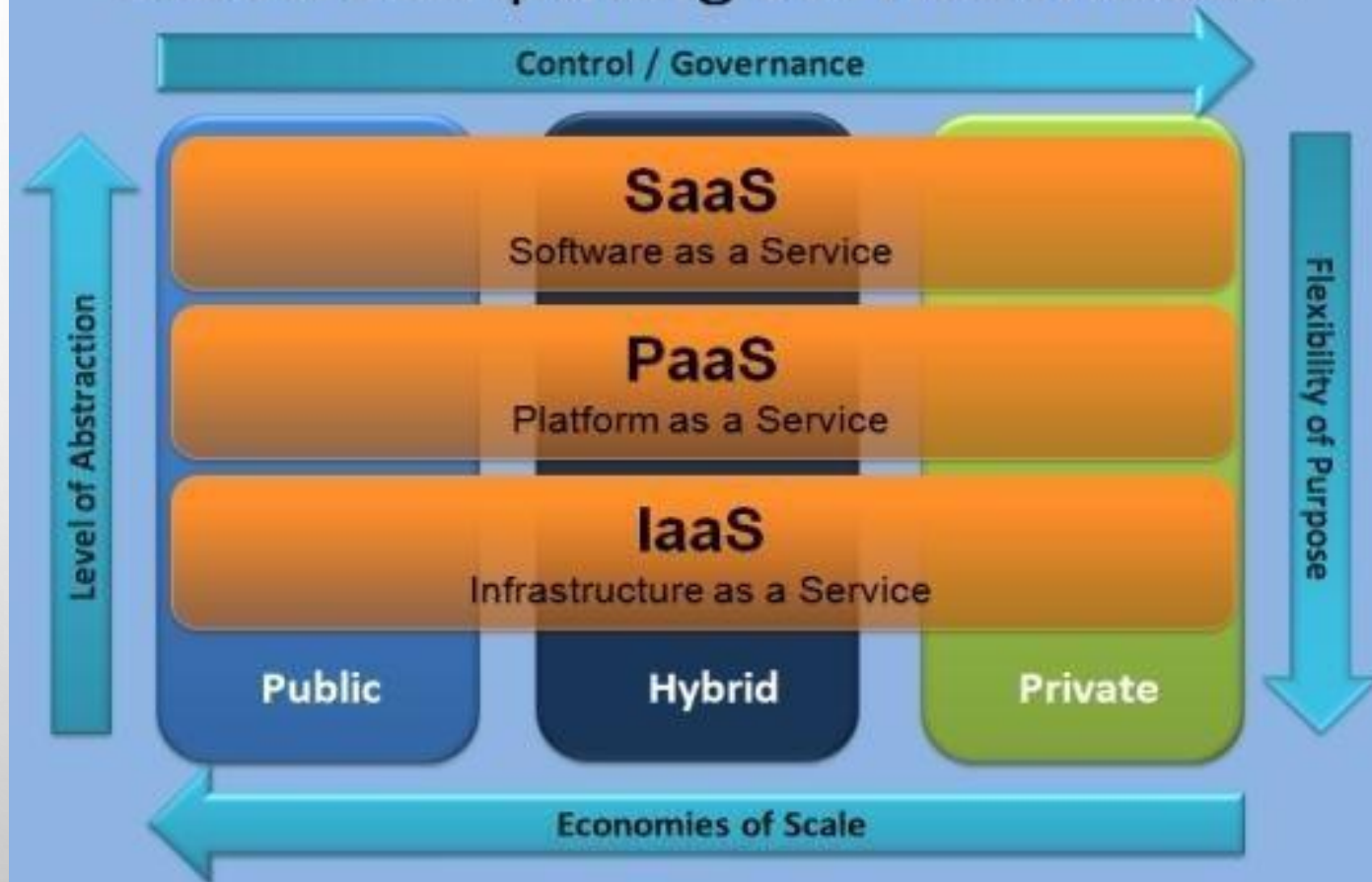
- **Infrastructure as a service (IaaS)**
 - IaaS provides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc.
 - IaaS examples: AWS EC2, Rackspace, Google Compute Engine (GCE), Digital Ocean, Magento 1 Enterprise Edition*.
- **Platform as a service (PaaS)**
 - PaaS provides the runtime environment for applications, development & deployment tools, etc.
 - PaaS examples: AWS Elastic Beanstalk, Heroku, Windows Azure (mostly used as PaaS), Force.com, OpenShift, Apache Stratos, Magento Commerce Cloud.
- **Software as a service (SaaS)**
 - SaaS model allows to use software applications as a service to end users.
 - SaaS examples: BigCommerce, Google Apps, Salesforce, Dropbox, MailChimp, ZenDesk, DocuSign, Slack, Hubspot.



SERVICE MODELS

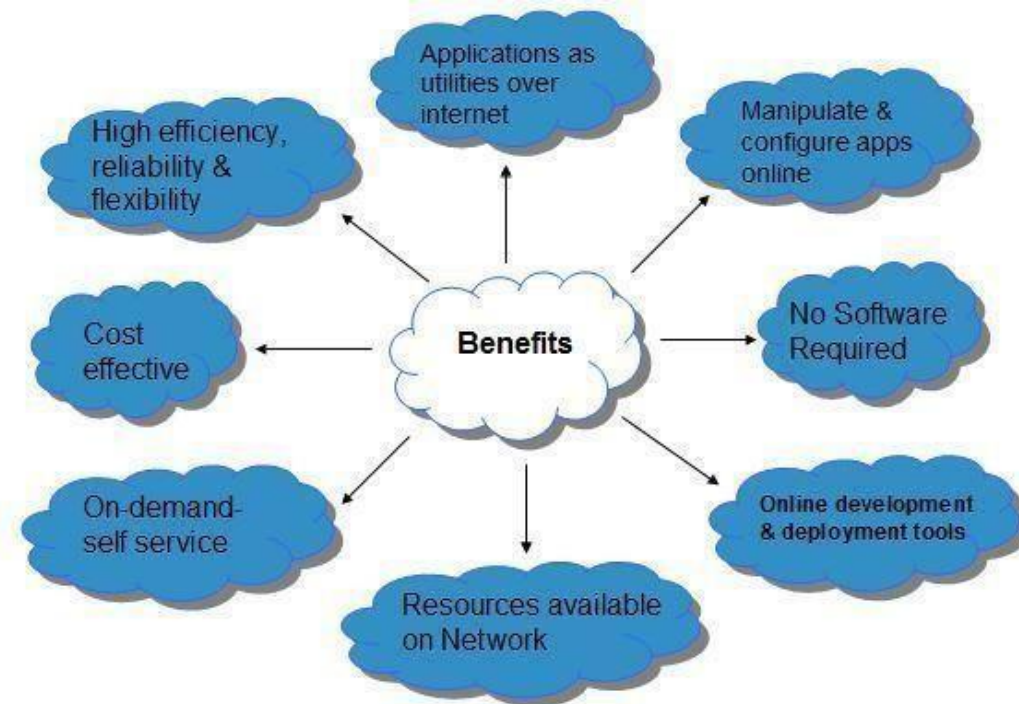
Model	Full Form	Description	Examples
IaaS	Infrastructure as a Service	Provides fundamental computing resources such as virtual machines, storage, and networks. Users install their own OS and apps.	AWS EC2, Google Compute Engine, Microsoft Azure VMs
PaaS	Platform as a Service	Provides a development environment with tools, libraries, and frameworks to build, test, and deploy applications.	Google App Engine, Microsoft Azure App Service, Heroku
SaaS	Software as a Service	Provides ready-to-use applications over the internet. The provider manages infrastructure and software.	Gmail, Salesforce, Google Docs, Microsoft 365

Cloud Computing Service Models



BENEFITS OF CLOUD COMPUTING

- **Access applications as utilities**, over the Internet
- **Manipulate and configure the application online** at any time
- It **does not require installation of a specific software** to access or manipulate cloud application.
- Offers **online development and deployment tools**, programming runtime environment through PaaS model.
- **Cloud resources are available over the network** in a manner that provides platform independent access to any type of clients.
- **Offers on-demand self-service**. The resources can be used without interaction with cloud service provider.
- It is highly **cost effective** because it operates at higher efficiencies with greater utilization.
- Offers **load balancing** that makes it more reliable



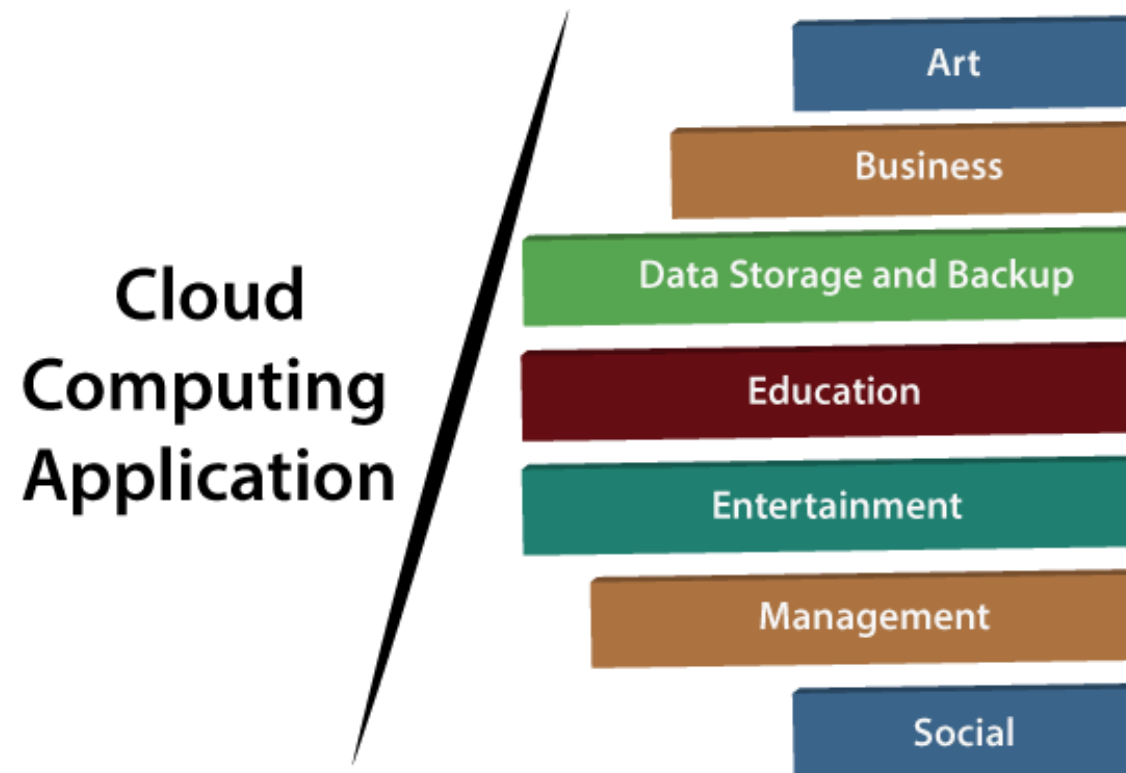
CHALLENGES OF CLOUD COMPUTING

- **Security and Privacy** - It is the biggest challenge to cloud computing, which can be overcome by employing encryption, security hardware and security applications.
- **Portability** - Each of the cloud provider uses different standard languages for their platforms and is difficult to migrate from one cloud provider to another.
- **Interoperability** - It means the application on one platform should be able to incorporate services from the other platforms.
- **Computing Performance** - Data intensive applications on cloud requires high network bandwidth, which results in high cost. Low bandwidth does not meet the desired computing performance of cloud application.
- **Reliability and Availability** - Businesses are now becoming dependent on services provided by third-party.



CLOUD COMPUTING APPLICATIONS

- Cloud Service Providers provide various applications in the field of art, business, data storage and backup services, education, entertainment, management, social networking, etc.
- The most widely used cloud computing applications are :-



CLOUD COMPUTING APPLICATIONS

Art Applications

- Cloud computing offers various art applications for quickly and easily design attractive cards, booklets, and images.
- Some most commonly used cloud art applications are given below:
 1. **Moo** - One of the best cloud art applications used for designing and printing business cards, postcards, and mini cards.
 2. **Vistaprint** - Allows us to easily design various printed marketing products such as business cards, Postcards, Booklets, and wedding invitations cards.
 3. **Adobe Creative Cloud** - Made for designers, artists, filmmakers, and other creative professionals. It is a suite of apps which includes PhotoShop image editing programming, Illustrator, InDesign, TypeKit, Dreamweaver, XD, and Audition.



CLOUD COMPUTING APPLICATIONS

Business Applications

- Today, every organization requires the cloud business application 24*7 to grow their business.
- There are the following business applications of cloud computing
 1. **MailChimp** - email publishing platform which provides various options to design, send, and save templates for emails.
 2. **Salesforce** - cloud development platform which provides tools for sales, service, marketing, e-commerce, and more.
 3. **Chatter** - share important information about the organization in real time
 4. **Bitrix24** - provides communication, management, and social collaboration tools
 5. **Paypal** - Paypal offers the simplest and easiest online payment mode using a secure internet account.
 6. **Slack** - Slack (Searchable Log of all Conversation and Knowledge) provides a user-friendly interface that helps us to create public and private channels for communication.
 7. **Quickbooks** - Quickbooks works on the terminology "Run Enterprise anytime, anywhere, on any device." It provides online accounting solutions for the business with more than 20 users to work simultaneously on the same system.



mailchimp



PayPal



slack



INTUIT
quickbooks

CLOUD COMPUTING APPLICATIONS

Data Storage and Backup Applications

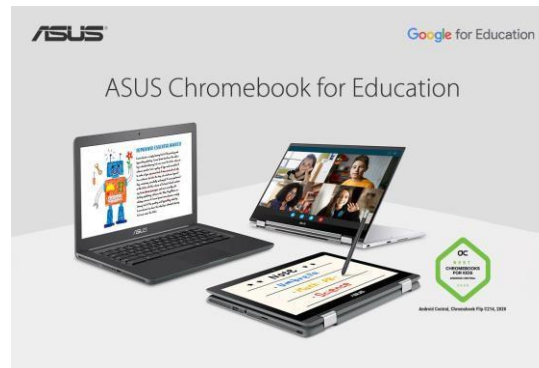
- Cloud computing allows us to store information (data, files, images, audios, and videos) and provide security by offering various backup recovery application for retrieving the lost data.
- A list of data storage and backup applications in the cloud: -
 1. **Box.com** - Provides an online environment for secure content management, workflow, and collaboration. Allows us to store different files such as Excel, Word, PDF, and images on the cloud and provides drag & drop service for files and easily integrates with Office 365, G Suite, Salesforce etc.
 2. **Joukuu** - Provides the simplest way to share and track cloud- based backup files, search files, folders, and collaborate on documents.
 3. **Google G Suite** - One of the best cloud storage and backup application the Google G Suite which includes Gmail, Google Calendar, Docs, Forms, Google+, Hangouts, as well as cloud storage and tools for managing cloud apps.



CLOUD COMPUTING APPLICATIONS

■ Education Applications

- It offers various online distance learning platforms and student information portals to the students.
- Offers strong virtual classroom environments, ease of accessibility, secure data storage, scalability, greater reach for the students, and minimal hardware requirements for the applications.
- There are the following education applications offered by the cloud -
 1. **Google Apps for Education** - The most widely used platform for free web-based email, calendar, documents, and collaborative study.
 2. **Chromebooks for Education** - Designed for the purpose that it enhances education innovation.
 3. **Tablets with Google Play for Education** - Allows educators to quickly implement the latest technology solutions into the classroom and make it available to their students.
 4. **AWS in Education** - Provides an education-friendly environment to universities, community colleges, and schools.



CLOUD COMPUTING APPLICATIONS

Entertainment Applications

- Entertainment industries use a multi-cloud strategy to interact with the target audience.
- Cloud computing offers various entertainment applications such as online games and video conferencing.
 1. **Online games** - Today, cloud gaming becomes one of the most important entertainment media. It offers various online games that run remotely from the cloud. The best cloud gaming services are GeForce Now, Vortex, Project xCloud, and PlayStation Now.
 2. **Video Conferencing Apps** - Video conferencing apps provides a simple and instant connected experience. It allows us to communicate with our business partners, friends, and relatives using a cloud-based video conferencing. The benefits of using video conferencing are that it reduces cost, increases efficiency, and removes interoperability.



CLOUD COMPUTING APPLICATIONS

Management Applications

- Cloud computing offers various cloud management tools which help admins to manage all types of cloud activities, such as resource deployment, data integration, and disaster recovery.
- Provide administrative control over the platforms, applications, and infrastructure.
- Some important management applications are -
 1. **Toggl** - Helps users to track allocated time period for a particular project.
 2. **Evernote** - Allows us to sync and save our recorded notes, typed notes, and other notes in one convenient place using platforms like Windows, macOS, Android, iOS, Browser, and Unix.
 3. **Outright** - It is used by management users for the purpose of accounts to track income, expenses, profits, and losses in real-time environment.
 4. **GoToMeeting** - Provides Video Conferencing and online meeting apps, which allows us to start a meeting with our business partners from anytime, anywhere using mobile phones or tablets and perform the tasks related to the management such as join meetings, view presentations on the shared screen, get alerts for upcoming meetings, etc.



CLOUD COMPUTING APPLICATIONS

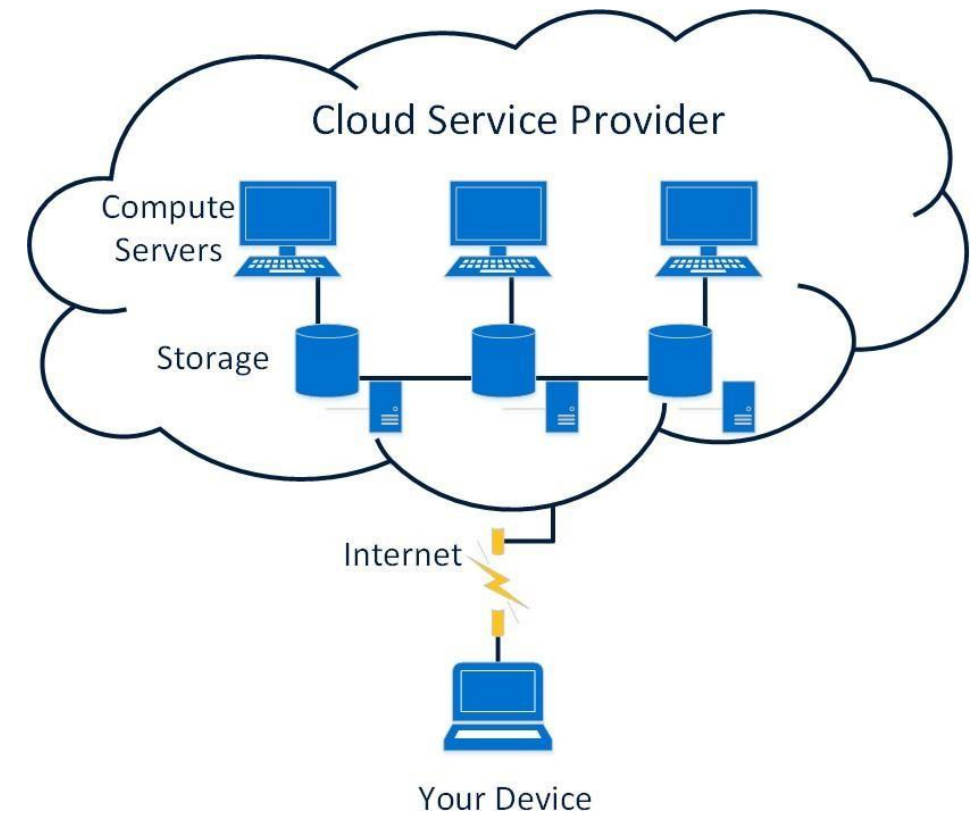
Social Applications

- Social cloud applications allow a large number of users to connect with each other using social networking applications such as Facebook, Twitter, LinkedIn, etc.
- There are the following cloud based social applications -
 1. **Facebook** - Social networking website/application which allows active users to share files, photos, videos, status, more to their friends, relatives, and business partners using the cloud storage system.
 2. **Twitter** - It is a social networking site and a microblogging system. It allows users to follow high profile celebrities, friends, relatives, and receive news by sending and receiving short posts called tweets.
 3. **Yammer** - Team collaboration tool that allows a team of employees to chat, share images, documents, and videos.
 4. **LinkedIn** - LinkedIn is a social network for students, freshers, and professionals.



CLOUD STORAGE

- Cloud storage is the process of storing digital data in an online space that spans multiple servers and locations, and it is usually maintained by a hosting company.
- Essentially, an individual or organization can store and access data in this online space maintained by a host service using the internet.
- You are likely using cloud storage already
 - whether it is Microsoft Office 365, Dropbox, Google Drive
 - or one of the other dozens of cloud storage providers.
- Even media sharing devices like Instagram and YouTube or webmail clients like Gmail and Hotmail fall into this category.
 - Each of these services stores data – our data – on the cloud.



BENEFITS OF CLOUD STORAGE

- **Cloud Data Retrieval:**
 - We can retrieve data from virtually anywhere.
 - Companies can now offer **work-from-home** and **bring-your-own-device (BYOD)** solutions.
- **Collaboration:**
 - Many people can access the same data and tools, enabling teams to work together.
- **Pay-As-You-Use:**
 - In the past, businesses had to purchase computing infrastructure – hardware, licenses, equipment – based on their expectations on business growth for the next several years. Companies tended to over-buy, fearing being underprepared, and were left with expensive equipment and storage space they didn't use.
 - Cloud storage and virtualization allows businesses to pay for the data they actually use and can scale and adjust up and down as their business needs and strategy change.
- **Increased Capabilities:**
 - Small and medium-sized businesses of the past were only able to use the tools they could afford. Today, using a cloud storage provider means SMBs can utilize the different services the cloud provider offers for all their clients.

CHALLENGES OF CLOUD STORAGE

- **Attack Surface Area:**

- This is a mathematical understanding of risk.
- By distributing data to more locations, the risk of that data being subject to unauthorized access increases. More people = more risk of compromise. The cloud also uses the internet to transfer data.
- So, instead of data being transferred via a local area network (LAN), it uses a wide area network (WAN), which increases the risk.

- **Supplier Sustainability:**

- What happens if the cloud storage provider goes bankrupt, suffers a disaster, or changes their business strategy?

- **Security:**

- Many businesses store sensitive data, such as healthcare or financial records which are subject to strict regulations.
- Putting this data in the hands of someone else can be a difficult decision to make.
- Understanding the safeguards cloud storage providers take and how our team will need to leverage those tools for added security is an important step in migrating our data to the cloud.

TYPES OF CLOUD STORAGE

1. Public Cloud

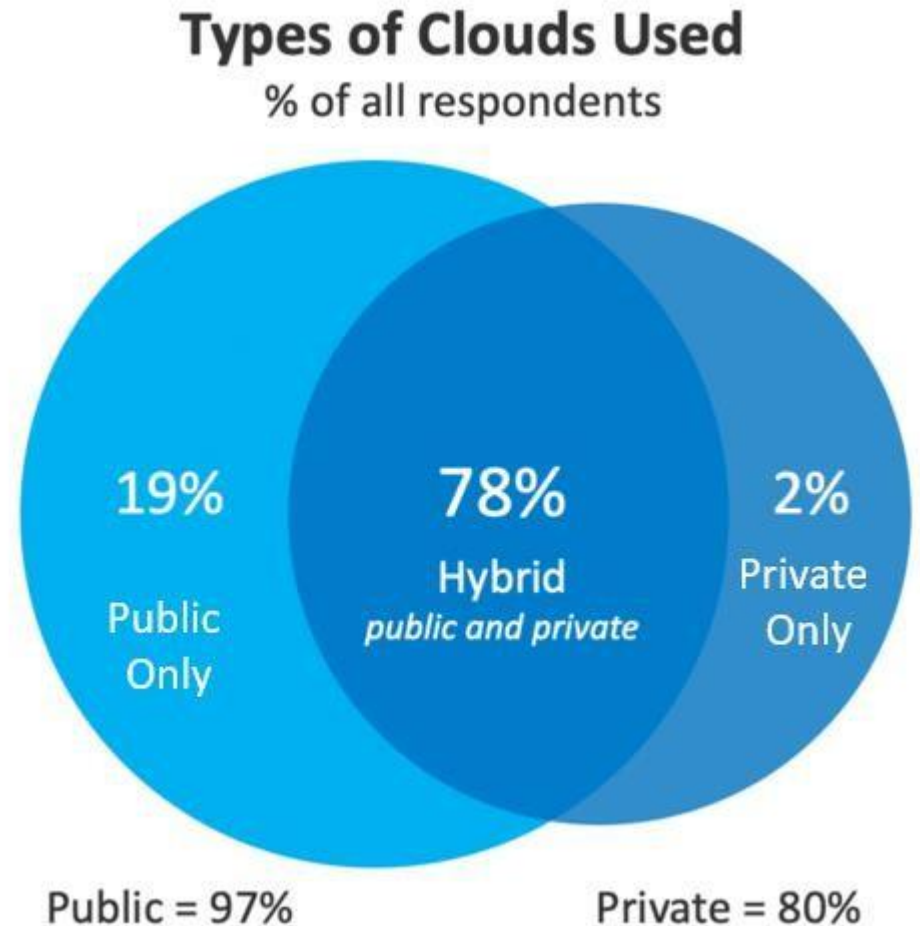
- Public cloud storage is basically compatible with unstructured data. It provides a multi-tenant storage environment.
- The data store in multiple data centers and can access across multiple regions for continents. The cloud storage providers completely manage the public cloud storage.

2. Private Cloud

- Private cloud is mostly compatible with the customers you need to customize their control over their data. Here, the enterprise and the cloud storage providers combine the data center to serve the customer better.

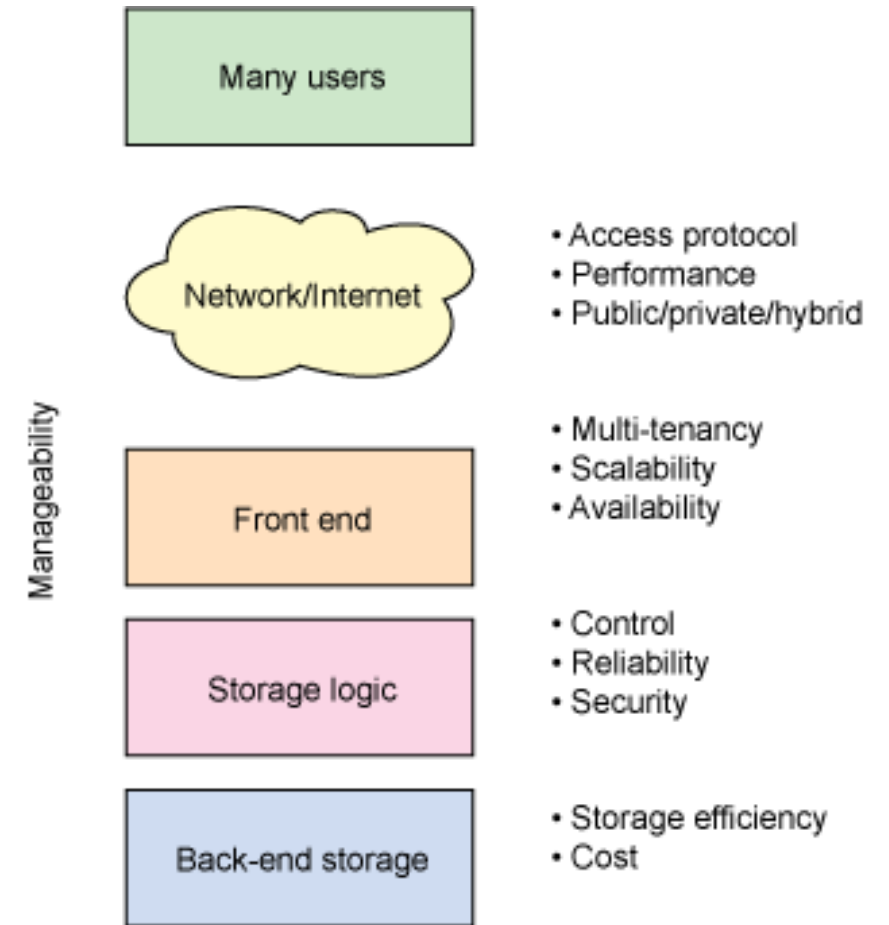
3. Hybrid Cloud

- Hybrid cloud is a combination of private and personal cloud which can modify as per the demand. It provides more flexibility to the customer and has more data deployment options.
- Hybrid cloud is suitable for both small and large sector organizations. It accessible from anywhere and can provide numerous amount of benefits the customer.



CLOUD STORAGE INFRASTRUCTURE

- Cloud storage has an infrastructure which bases on the **principle of virtualization** in Cloud Computing.
- **Virtualization** eliminates the hardware by utilizing the single hardware and making many virtual separations of it.
- Its infrastructure is **elastic**, **scalable**, and **multi-tenant**. With the help of proper tools, the whole infrastructure can manage and information can store and retrieve easily.
- With the help of **ID** and **password**, the whole content can retrieve from anywhere and at any time.
- In addition, there are proper **security measures** which authenticate the whole process of uploading and downloading.



Generic Cloud Storage Architecture

RISK RELATED TO CLOUD STORAGE

- Security & Privacy

Since data management and infrastructure management in cloud is provided by third-party, it is always a risk to handover the sensitive information to such providers. Any sign of security breach would result in loss of clients and businesses.

- Lock-in

It is very difficult for the customers to switch from one Cloud Service Provider (CSP) to another.

- Isolation failure

This risk involves the failure of isolation mechanism that separates storage, memory, routing between the different tenants.

- Management interface compromise

In case of public cloud provider, the customer management interfaces are accessible through the Internet.

- Insecure or incomplete data deletion

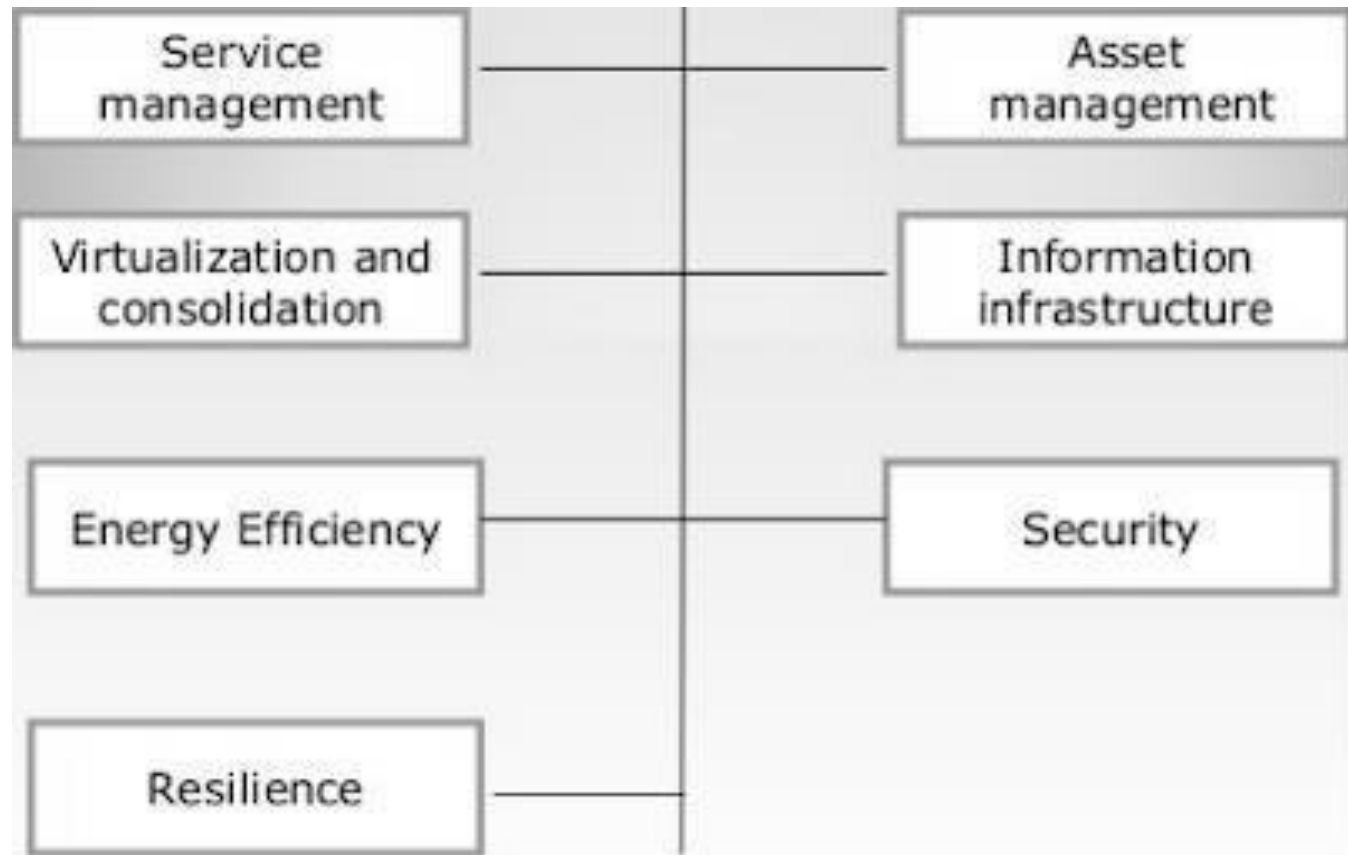
It is possible that the data requested for deletion may not get deleted. It happens either because extra copies of data are stored but are not available or disk destroyed also stores data from other tenants.

CLOUD SERVICE REQUIREMENT

- **Efficiency / Cost Reduction:** By minimizing expenses in purchasing and maintaining equipment.
- **Data Security:** Cloud storage providers implement baseline protections for their platforms and the data they process, such authentication, access control, and encryption.
- **Scalability:** It enables enterprise to efficiently -- and quickly -- scale up/down according to business demands.
- **Mobility:** Cloud computing allows mobile access to corporate data via smartphones and devices, which is a great way to ensure that no one is ever left out of the loop.
- **Disaster Recovery:** Data loss is a major concern for all organizations, along with data security. Storing our data in the cloud guarantees that data is always available.
- **Control:** Cloud enables you complete visibility and control over our data. We can easily decide which users have what level of access to what data.
- **Market reach:** Developing in the cloud enables users to get their applications to market quickly.
- **Automatic Software Updates:** Cloud-based applications automatically refresh and update themselves.

CLOUD AND DYNAMIC INFRASTRUCTURE

1. Service Management
2. Asset-Management
3. Virtualization and Consolidation
4. Information Infrastructure
5. Energy-Efficiency
6. Security
7. Resilience



CLOUD AND DYNAMIC INFRASTRUCTURE

1. **Service Management:** Includes visibility, automation and control to delivering the first-class IT services.
2. **Asset-Management:** In this the assets or the property which is involved in providing the cloud services are getting managed.
3. **Virtualization and Consolidation:** Consolidation is an effort to reduce the cost of a technology by improving its operating efficiency and effectiveness. It means migrating from large number of resources to fewer one, which is done by virtualization technology.
4. **Information Infrastructure:** It helps the business organizations to achieve the following: Information compliance, availability of resources retention and security objectives.
5. **Energy-Efficiency:** Here the IT infrastructure or organization sustainable. It means it is not likely to damage or effect any other thing.
6. **Security:** This cloud infrastructure is responsible for the risk management. Risk management Refers to the risks involved in the services which are being provided by the cloud-service providers.
7. **Resilience:** This infrastructure provides the feature of resilience means the services are resilient. It means the infrastructure is safe from all sides. The IT operations will not be easily get affected.

CLOUD ADOPTION

- Cloud adoption is a strategy used by enterprises to improve the scalability of Internet-based database capabilities while reducing cost and risk.
- To achieve this, businesses engage in the practice of cloud computing or using remote servers hosted on the Internet to store, manage, and process critical data.
- While cloud computing has been available to the general public for several years, hybrid cloud computing is a relatively newer concept combining one or more cloud providers, such as Amazon Web Services, SAP HANA Cloud Platform, VMWare, or Salesforce, with a **private IT infrastructure** designed for a **specific organization**.



CLOUD ADOPTION

Who Needs Cloud Adoption?

1. Healthcare:

- Fueled by digital and social consumer behaviors and the need for secure and accessible electronic health records (EHRs), hospitals, clinics, and other medical organizations are using cloud computing for document storage, marketing, and human resources.

2. Marketing and Advertising:

- In an industry dependent on social media, as well as the quick creation and publishing of customer-relevant content, agencies are using hybrid cloud adoption strategies to deliver critical client messages to their local and worldwide audiences.

3. Retail:

- Internet-based retail is able to effectively market to customers and save their product data for less money.

4. Finance:

- Financial services institutions are now placing their email platforms and marketing tools in the cloud for efficient expense management, human resources, and customer communications are three of the most important business needs of today's finance organizations.

5. Education:

- The cloud allows universities, private institutions, and K-12 public schools to provide learning, homework, and grading systems online.

THANK YOU !